

≡ Hide menu

Module 3 Introduction

Lesson 5: Robust Regression

Lesson 6: Model Validations

Module 3 Summative Assessment

✔

 Graded Assignment: Module 3 Summative Assessment

Submitted

Module 3 Summary

Module 3 Summative Assessment

Review Learning Objectives

ⓘ

You ran out of attempts. If you'd like to try again, you will have attempts available in 7h 51m

Assignment details

Due

Feb 4, 11:59 PM IST

Attempts

0 left (1 attempts every 8 hours)

Submitted

Jan 23, 3:32 AM IST

Your grade

To pass you need at least 80%. We keep your highest score.

100%

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

Instructions

1. In the k -fold cross-validation technique, the data are shuffled randomly and then divided into k equal sub-samples. Which of the following statement(s) is/are true?

1 / 1 point

(I) We then use all the samples as training samples to be learned from.

(II) We then use $k - 1$ samples as training samples and the k -th sample as a test sample.

(III) We then use $k - 2$ samples as training samples, and 2 samples as test samples.

☐ I only

☒ II only

☐ III only

☐ II and III only

☐ I, II, and III

✔

 Correct

2. Which of the following R -command is/are related to setting up a robust regression using the MASS library?

1 / 1 point

(I) `rlm(y ~ x, data = df, psi = psi.bisquare)`

(II) `rlm(y ~ x1 + x2, data = df, psi = psi.huber)`

(III) `glm(y ~ x, data = df, family = binomial)`

☐ I only

☐ II only

☐ III only

☒ I and II only

☐ I, II, and III

✔

 Correct

3. Which of the following can be used to measure how good a model can predict the actual observations.

1 / 1 point

(I) A high R^2 score.

(II) A high MAE score.

(III) A high RMSE score.

☒ I only

☐ II only

☐ III only

☐ II and III only

☐ I, II, and III

✔

 Correct

4. Which of the following is an advantage of using the M -estimation technique?

1 / 1 point

To detect and remediate multicollinearity issue in the data.

To obtain estimations of the regression parameters using the least square method.

To detect and delete outliers in the data.

M -estimation is used as a variance stabilizing technique.

M -estimation is a robust regression technique used when the data has outliers.

✔

 Correct

5. Which of the following can be used in the variable selection process?

1 / 1 point

Ridge regression

Forward stepwise regression

Backward stepwise regression

☐ I only

☐ II only

☐ III only

☒ II and III only

☐ I, II, and III

✔

 Correct

👍 Like

👎 Dislike

🚩 Report an issue

View submission

See feedback

Try again

